

10 Mark(SPGD)

Q.1 Describe the construction details & operation of Buchholz relay.

Ans. 2. Construction of Buchholz Relay The relay is a gas-actuated device placed in the oil pipeline between transformer tank and conservator.

Main Parts

(i) Relay Chamber

- .A metallic chamber filled with transformer oil
- .Connected between tank and conservator
- .Acts as housing for internal components

(ii) Float System

There are two floats inside:

Upper Float (Alarm Float)

Positioned at the top

Operates during minor faults

Connected to alarm circuit

.Lower Float (Trip Float)

Located at the bottom

Operates during serious faults

Connected to tripping circuit

(iii) Mercury Switches

Attached to both floats

When float tilts → mercury flows → completes electrical circuit

(iv) Gas Collection Chamber

Located at top of relay

Stores gas produced due to internal faults

Gas accumulation lowers oil level

(v) Oil Flow Arrangement

Relay has a passage for oil flow between tank and conservator

Sudden oil movement operates lower float

3. Operation of Buchholz Relay

The operation depends on **type and severity of internal fault**.

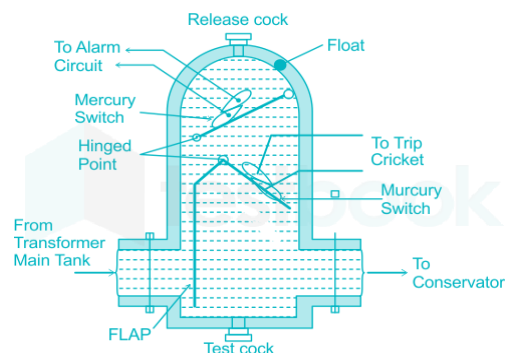
(A) Operation During Minor Faults

Examples

Insulation deterioration

Inter-turn fault

Core heating



Q.2 Explain transformer protection using differential protection method (Merz-Price Protection).

Ans. 2. Basic Principle Under normal conditions:

Current entering primary = Current leaving secondary (after ratio adjustment)

$$I_1 = I_2$$

👉 No current flows through relay → No operation

Currents become unequal

$$I_1 \neq I_2$$

👉 Difference current flows through relay → Relay operates → Circuit breaker trips

3. Construction

Main Components

Current Transformers (CTs) on both primary and secondary sides

Pilot wires connecting CTs

Differential relay

Circuit breakers

Connection (Circulating Current Scheme)

CT secondaries are connected in parallel (opposition)

Relay is placed between them

4. Operation

(A) Normal Condition

Load current flows through transformer

CTs produce proportional currents

These currents circulate between CTs

No current flows through relay 👉 **Relay remains inactive**

(B) External Fault (Outside Protection Zone)

Fault occurs outside transformer

(C) Internal Fault (Inside Protection Zone)

Differential current flows through relay 2 Relay operates

Circuit breakers trip → transformer isolated

5. Important Features

(i) Zone of Protection

Only protects the section between the two CTs

Called protected zone

(ii) Percentage Differential Protection

Used to avoid false tripping due to:

CT errors

Magnetizing inrush current

Relay operates only if:

Differential current $>$ bias value

(iii) Harmonic Restraint

Prevents operation during inrush current

Inrush current contains harmonics $\rightarrow$ relay blocks operation

6. Advantages

Very fast operation

High accuracy and sensitivity

Selective protection (only faulty zone isolated)

Stable during external faults

7. Disadvantages

Requires matched CTs

Sensitive to CT errors and saturation

More costly and complex

Not economical for small transformers

8. Applications

Power transformers

Generators

Bus bars

Large motors

9. Conclusion

The Merz-Price differential protection system is a highly effective method for transformer protection. By continuously comparing input and output currents, it ensures quick detection of internal faults, minimizes damage, and improves system reliability.

Q.3 Describe the construction details & operation of Vacuum Circuit Breaker With merit and demerit.

Ans. . The main part of a VCB is the vacuum interrupter. Main Components

(i) Vacuum Interrupter Chamber

Completely sealed enclosure (vacuum $\approx 10^{-6}$ to 10^{-7} torr)

Made of ceramic or glass

Contains contacts and arc control elements

(ii) Fixed and Moving Contacts

Fixed contact $\rightarrow$ stationary

Moving contact $\rightarrow$ connected to operating mechanism

Made of special alloys (Cu–Cr) for high conductivity and arc resistance

(iii) Metallic Bellows

Provides airtight sealing

Allows movement of the moving contact without breaking vacuum

(iv) Arc Shield

Surrounds contacts

Prevents metal vapor deposition on enclosure

Maintains dielectric strength

(v) Operating Mechanism

Spring or motor-operated mechanism

Used to open/close contacts

3. Operation of VCB

(A) Closing Operation

Moving contact touches fixed contact

Circuit becomes complete $\rightarrow$ current flows normally

(B) Opening Operation

Contacts separate when fault occurs

Arc is formed due to metal vapor (not gas)

Since vacuum has:

(C) Arc Extinction

At current zero, metal vapour condenses quickly

Vacuum regains dielectric strength rapidly

Arc extinguishes instantly

Important Point

👉 Dielectric strength of vacuum recovers faster than any other medium

Q.4 The plant capacity of a 3-phase generating station consists of two 10,000 KVA generators of reactance 12 % each and one 5000KVA generator of reactance 18%. The generators are connected to the station bus-bars from which load is taken through three 5000KVA step-up transformers each having a reactance of 5%. Determine the maximum fault MVA which the circuit breakers on (i) low voltage side and (ii) high voltage side may have to deal with.

Ans. Given Data

Generators: $2 \times 10,000 \text{ kVA}$, $X = 12\%$

$1 \times 5,000 \text{ kVA}$, $X = 18\%$

Transformers: $3 \times 5,000 \text{ kVA}$, $X = 5\%$

Step 1: Choose Base

Take base = 10,000 kVA

Step 2: Convert Reactances to Common Base

Generators

Two generators:

Already on 10,000 kVA base $\rightarrow X = 12\%$

Third generator (5000 kVA $\rightarrow$ 10,000 kVA base):

$$X = 18 \times \frac{10000}{5000} = 36\%$$

Step 3: Equivalent Reactance of Generators (Parallel) $\frac{1}{X_{eq}} = \frac{1}{12} + \frac{1}{12} + \frac{1}{36}$

$$\frac{1}{X_{eq}} = \frac{3 + 3 + 1}{36} = \frac{7}{36}$$

$$X_{eq} = \frac{36}{7} = 5.14\%$$

Step 4: Fault MVA on Low Voltage Side $\text{Fault MVA} = \frac{\text{Base}}{X_{eq}} = \frac{10000}{0.0514} \approx$

194.6 MVA

👉 **Answer (i): LV side = 194.6 MVA**

Step 5: Transformer Reactance Conversion

Each transformer: $X = 5\%$ on 5000 kVA

Convert to 10,000 kVA base: $X = 5 \times \frac{10000}{5000} = 10\%$

Step 6: Equivalent Transformer Reactance (Parallel) $X_T = \frac{10}{3} = 3.33\%$

Step 7: Total Reactance for HV Side $X_{total} = 5.14 + 3.33 = 8.47\%$

Step 8: Fault MVA on High Voltage Side

$$\text{Fault MVA} = \frac{10000}{0.0847} \approx 118.1 \text{ MVA}$$

$$\text{Fault MVA} = \frac{10000}{0.0847} \approx 118.1 \text{ MVA}$$

Answer (ii): HV side = 118.1 MVA

Q.5 Write short note on: Static Relay and IDMT Relay.

Ans. 1. Static Relay

Definition A static relay is a protective relay that uses electronic components (like diodes, transistors, ICs) instead of moving mechanical parts to detect faults.

Construction

Measuring unit (electronic circuits)

Comparator unit

Output unit (tripping circuit)

Power supply

Working Principle

Electrical quantities (voltage/current) are converted into electrical signals

These signals are processed electronically

When the value exceeds a preset limit → relay sends trip signal

Features

No moving parts

Operates using solid-state devices

Advantages

Very fast operation

High accuracy and sensitivity

Low maintenance

Disadvantages

Requires auxiliary power supply

Sensitive to temperature and voltage variations

2. IDMT Relay (Inverse Definite Minimum Time Relay)

Definition

An IDMT relay is an overcurrent relay in which:

Operating time is inversely proportional to fault current

But has a minimum definite time limit

 **Key Difference (Quick View)**

Feature	Static Relay	IDMT Relay
Type	Electronic relay	Overcurrent relay
Working	Signal processing	Time-current characteristic
Speed	Very fast	Moderate
Usage	Advanced protection Feeder coordination	

Q.6 Define lightning arrestor and describe types of lightning arrestor.

Ans. A lightning arrestor is a protective device used in power systems to protect electrical equipment from high voltage surges caused by lightning.

2. Working Principle (Brief)

Under normal voltage → acts as an insulator

During surge → becomes a conductor

Provides a low-resistance path to ground

After surge → returns to insulating state

3. Types of Lightning Arrestors

(i) Rod Gap Arrestor

Construction

Two metal rods separated by an air gap

Working

During high voltage → spark occurs across gap

Surge current flows to ground

Features

Simple and cheap

Poor reliability

Horn Gap Arrestor

Working

Arc forms during surge

Heated air makes arc rise upward → arc extinguishes

Features

Better than rod gap

Used in early systems

(iii) Expulsion Type Arrestor

Fiber tube with internal electrode

Spark gap inside

Working

Arc produces gases

Gas pressure blows out arc → extinguishes it

Features

Good performance

Used in distribution systems

(iv) Valve Type Arrestor

Working

During surge → gaps break down

Resistors limit current

After surge → gaps restore insulation

Features

Reliable and widely used (older substations)

(v) Metal Oxide (ZnO) Arrestor (Modern Type)

Working

Highly non-linear resistance

Conducts during surge

Blocks current during normal condition

Q.7 Describe the construction & principle of operation of an induction type directional over current relay with a neat diagram.

Ans. 2. Construction The relay consists of two main elements:

(A) Directional Unit

Components

Aluminium disc (rotating type)

Two electromagnets: 1 Current coil (series magnet) Voltage coil (shunt magnet)

3 Control spring **4** Damping system

Function

Determines the direction of power flow

(B) Overcurrent Unit

Components 1 Induction disc 2 Current coil 3 Time delay mechanism (for IDMT characteristics)

Function 1 Operates when current exceeds preset value

(C) Other Parts 1 Spindle and moving contacts 2 Fixed contacts 3 Trip circuit connection

Basic Principle

It works on electromagnetic induction and torque production.

Torque Equation

$$T \propto VI \sin \theta$$

Where:

V = voltage

I = current

θ = phase angle between V and I

Step 1: Direction Check

Current coil and voltage coil produce magnetic fluxes

These fluxes interact and produce torque

Torque direction depends on power flow direction

Step 2: Overcurrent Check

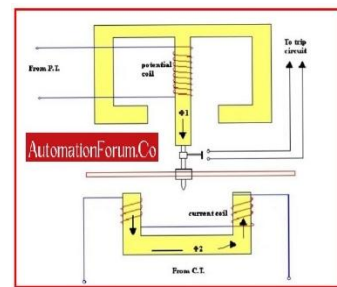
If current exceeds pickup value → overcurrent unit activates

Time delay may be applied (IDMT)

Step 3: Tripping Action

Disc rotates → closes contacts

Circuit breaker trips → faulty section isolated



5. Operation Conditions

Condition	Relay Action
Normal condition	No operation
Reverse direction fault	No operation
Forward fault but low current	No operation
Forward fault + high current	Relay operates

6. Advantages

- Provides directional discrimination
- Avoids unnecessary tripping

7. Disadvantages

- More complex than simple overcurrent relay
- Requires both current and voltage inputs

Q.8 Give a brief classification of circuit breakers. Discuss the construction details and operation of low oil circuit breaker.

Ans. 1. Classification of Circuit Breakers (Brief)

Circuit breakers are classified mainly based on the arc extinguishing medium:

(A) Based on Arc Quenching Medium

Oil Circuit Breaker

Bulk Oil CB

Low Oil CB

Air Circuit Breaker (ACB)

SF₆ Circuit Breaker

Vacuum Circuit Breaker (VCB)

(B) Based on Voltage Level

Low voltage CB

Medium voltage CB

High voltage CB

(C) Based on Operating Mechanism

Manual operated 2 Spring operated 3 Pneumatic operated 4 Hydraulic operated

2. Low Oil Circuit Breaker (Minimum Oil CB)

3. Construction of Low Oil Circuit Breaker Unlike bulk oil CB, it uses small quantity of oil only for arc extinction.

Main Parts

(i) Interrupting Chamber

Contains fixed and moving contacts

Filled with small amount of oil

(ii) Contacts

Made of copper alloys 2 Includes: 3 Fixed contact 4 Moving contact

(iii) Arc Control Device

Special chamber around contacts 2 Controls and directs arc

(iv) Insulating Structure

Porcelain or insulating material 2 Separates live parts from earth

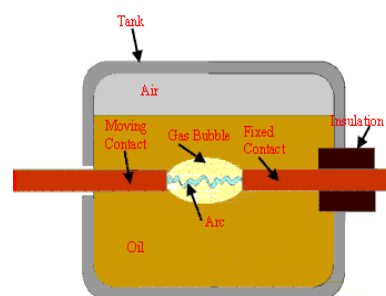
(v) Operating Mechanism

Spring or manual mechanism 2 Used to open and close contacts

4. Operation of Low Oil Circuit Breaker

(A) Normal Condition

Contacts remain closed 2 Current flows normally



Conceptual view of Bulk Oil Circuit Breaker

Q.9 Define lightning. What are the types of lightning strokes. Also state the harmful effects of lightning.

Ans. 1. Definition of Lightning Lightning is a sudden and violent electrical discharge between:

Cloud to earth, or Between clouds

It occurs due to the buildup of high electrostatic charges in the atmosphere during storms.

2. Types of Lightning Strokes Lightning strokes are classified based on how and where the discharge occurs:

(i) Direct Stroke

Lightning directly strikes: 1 Transmission lines 2 Towers 3 Substations

(ii) Indirect Stroke

Lightning strikes nearby ground or object 1 Induces high voltage surges in lines

(iii) Shielding Failure Stroke

1 Occurs when lightning bypasses shield wire 2 Hits phase conductor directly

(iv) Back Flashover

1 Lightning strikes tower or ground wire 2 High voltage causes flashover across insulator

(v) Cloud-to-Cloud Lightning

Discharge between two clouds 2 Does not directly affect system but can induce surges

3. Harmful Effects of Lightning

(i) Damage to Equipment

Breaks down insulation 2 Damages transformers, generators, and lines

(ii) Overvoltage in System

Produces very high transient voltages 2 Leads to system instability

(iii) Flashover of Insulators

Causes insulation failure 2 Leads to short circuits

(iv) Fire Hazards

Sparks may ignite fire in substations or buildings

(v) Power Interruption

Causes tripping of circuit breakers 2 Leads to blackout or supply interruption

(vi) Mechanical Damage

Can damage towers, poles, and structures

Q.10 Explain Merz-price voltage balance system for protection of feeders.

Ans. 2. Basic Principle

Under normal conditions:

Voltages at both ends of the feeder are equal

$$V_1 = V_2$$

No current flows through the relay → No operation

- During fault (within protected zone):

$$V_1 \neq V_2$$

Voltage difference produces current in pilot wires → Relay operates → Circuit breaker trips

3. Construction

Main Components

Two identical voltage transformers (VTs) at both ends
2 Pilot wires connecting both ends
3 Relays at each end
4 Circuit breakers

4. Operation

(A) Normal Condition

Voltages at both ends are equal
2 No current flows in pilot wires
3 Relay remains inactive

(B) External Fault

Fault outside protected zone

Working:

Voltage difference is negligible
2 Relay does not operate

(C) Internal Fault (Feeder Fault)

Working:

Fault causes drop in voltage at one end
2 Voltage difference appears between two ends
3 Current flows through pilot wires
4 Relay operates
5 Circuit breakers at both ends trip

5. Features

- Provides unit protection (only protected section is isolated)
- Fast and reliable
- Requires pilot wires